

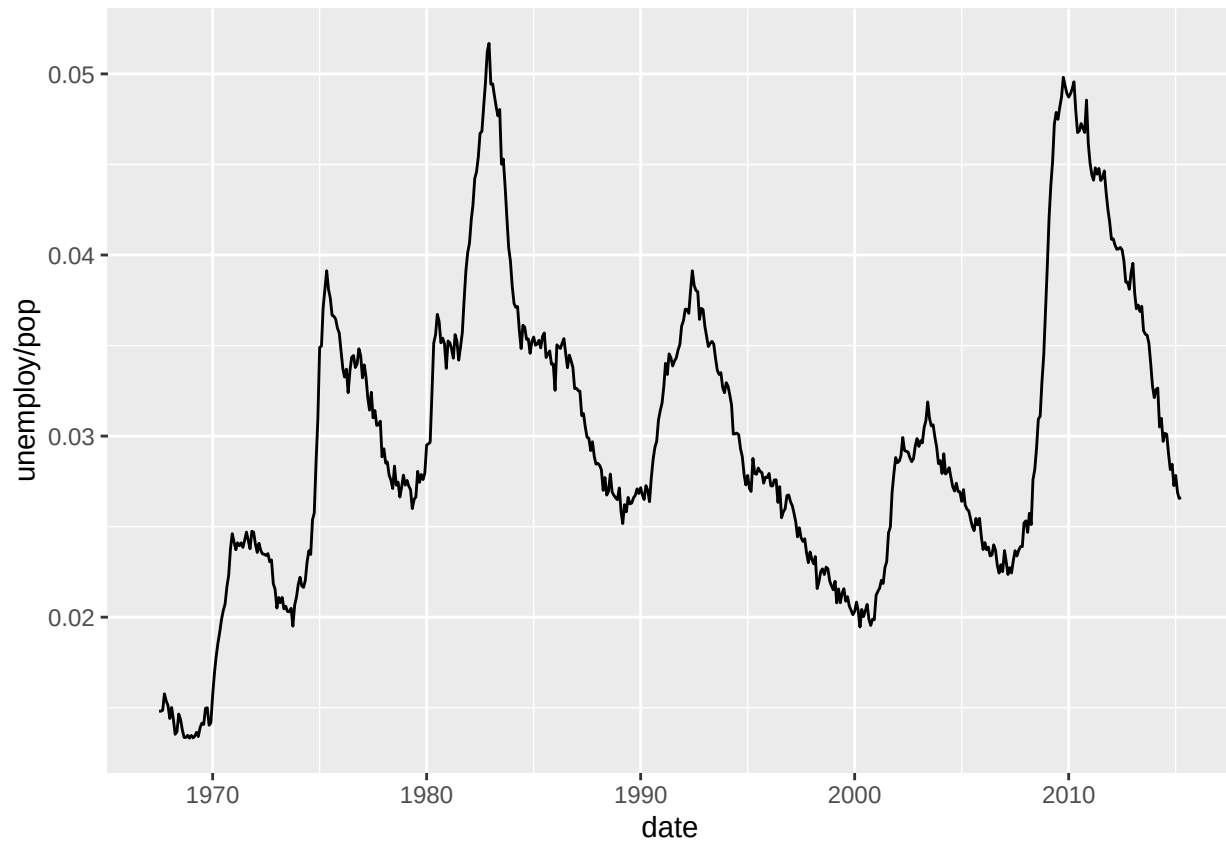
ex3-A113290007

Fabby

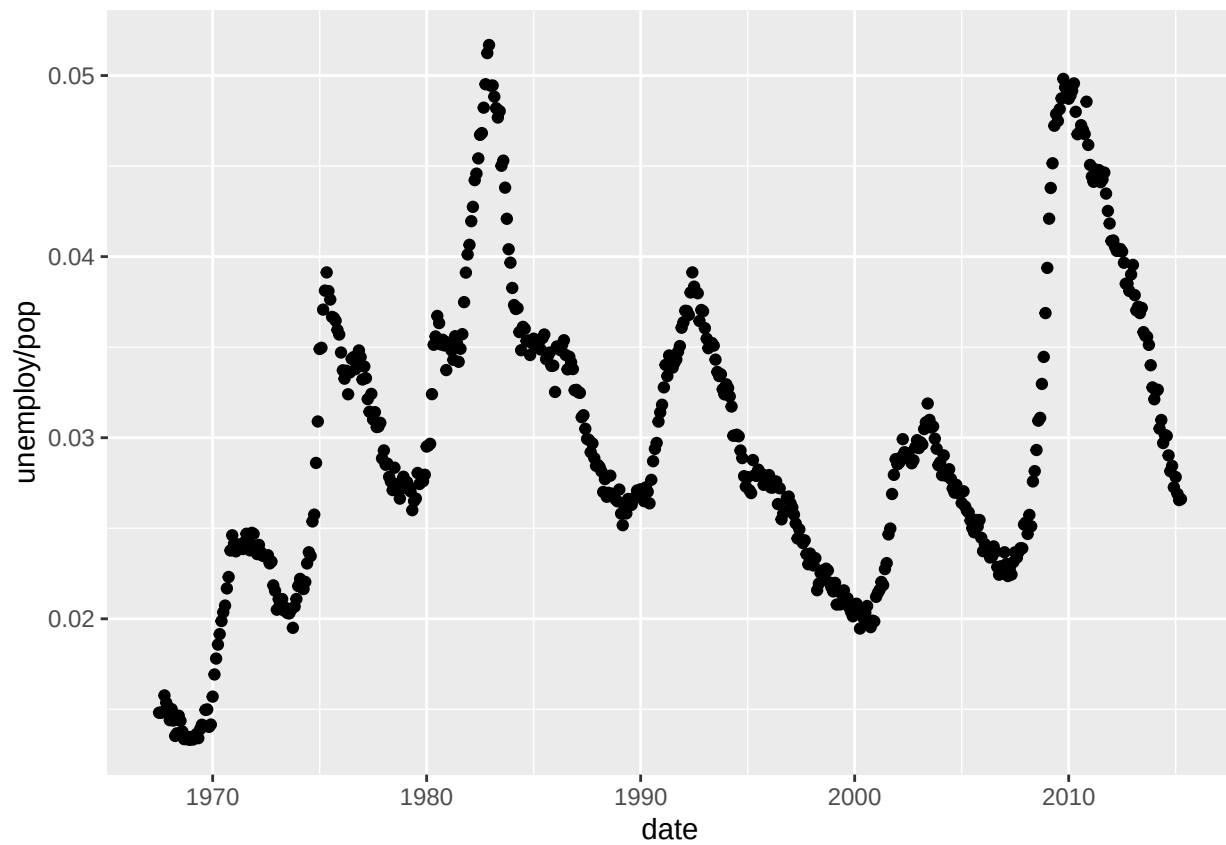
2026-10-07

q1

```
library(ggplot2)
ggplot(economics, aes(x = date, y = unemploy/pop)) + geom_line()
```

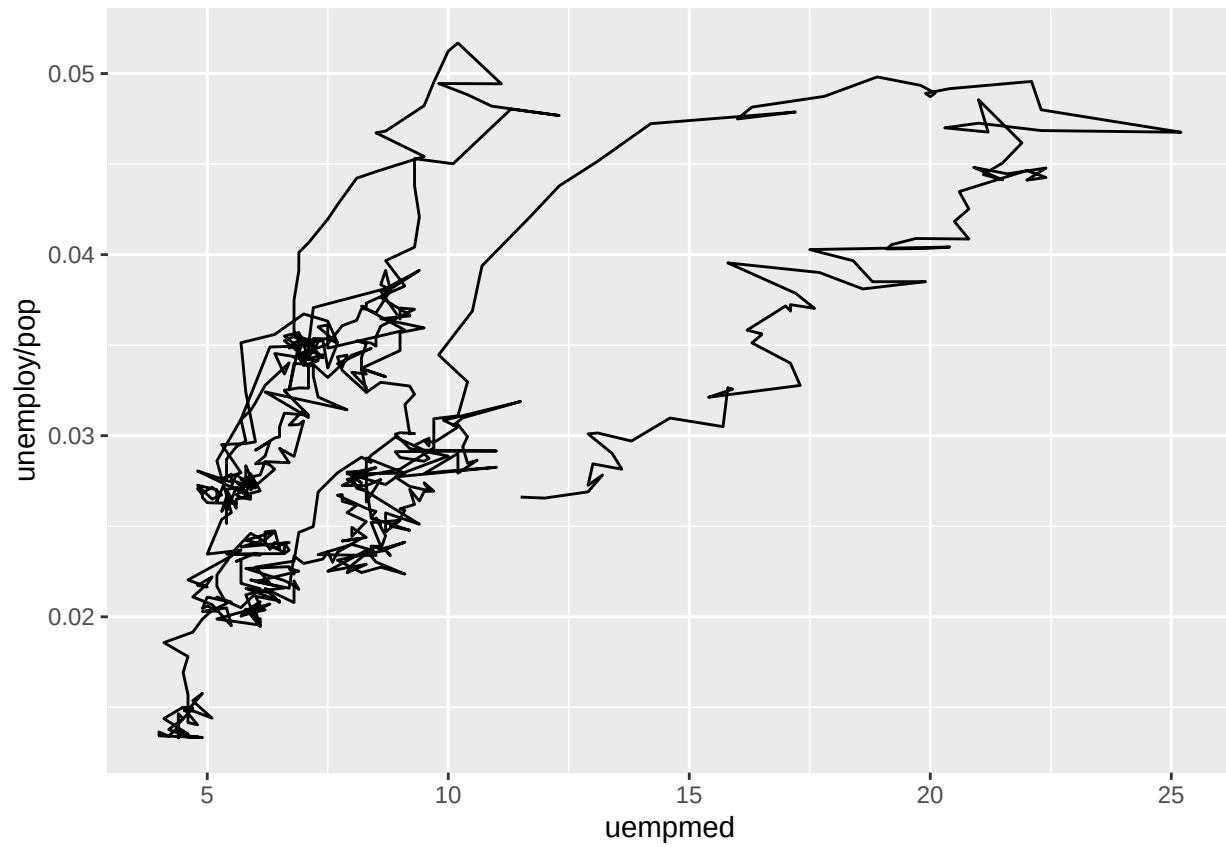


```
ggplot(economics, aes(x = date, y = unemploy/pop)) + geom_point()
```

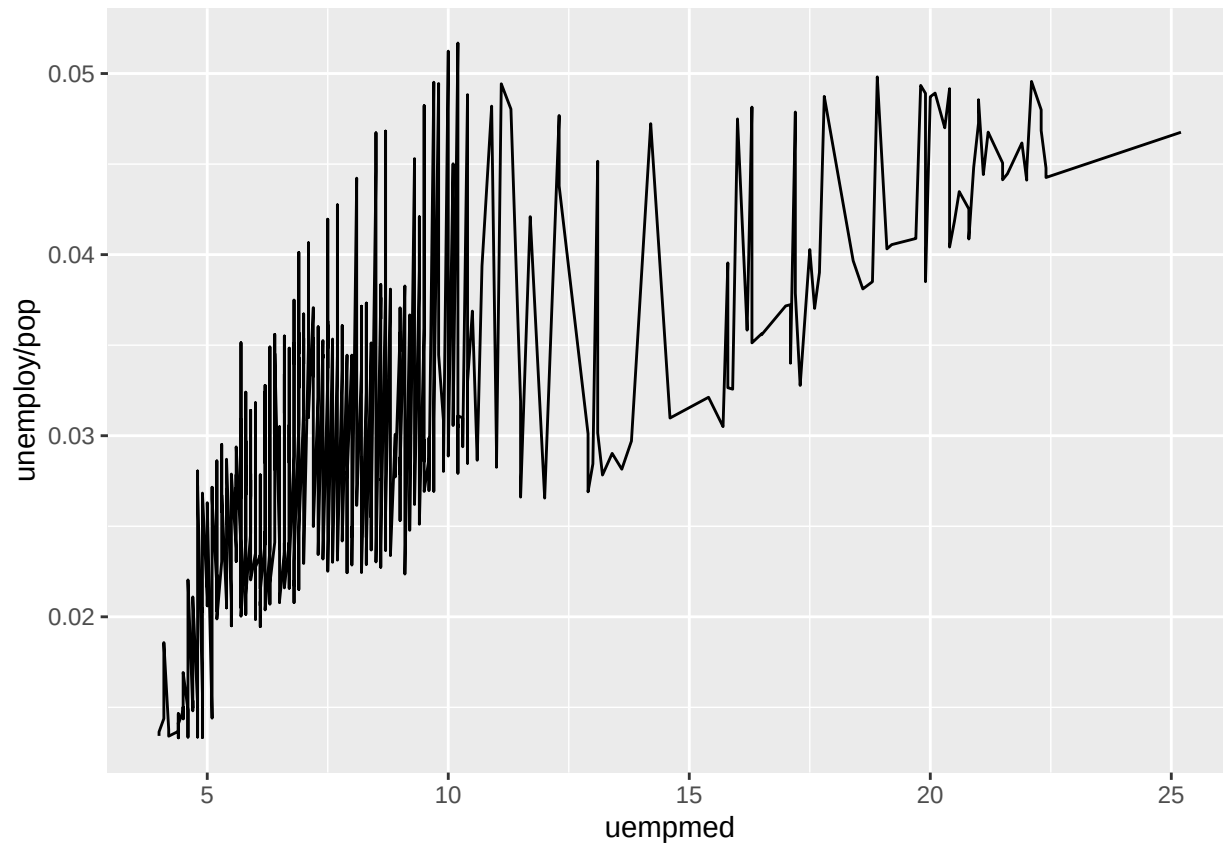


#q2

```
library(ggplot2)
# geom_path()
ggplot(economics, aes(x = uempmed, y = unemploy / pop)) +
  geom_path()
```



```
# geom_line()
ggplot(economics, aes(x = uempmed, y = unemploy / pop)) +
  geom_line()
```



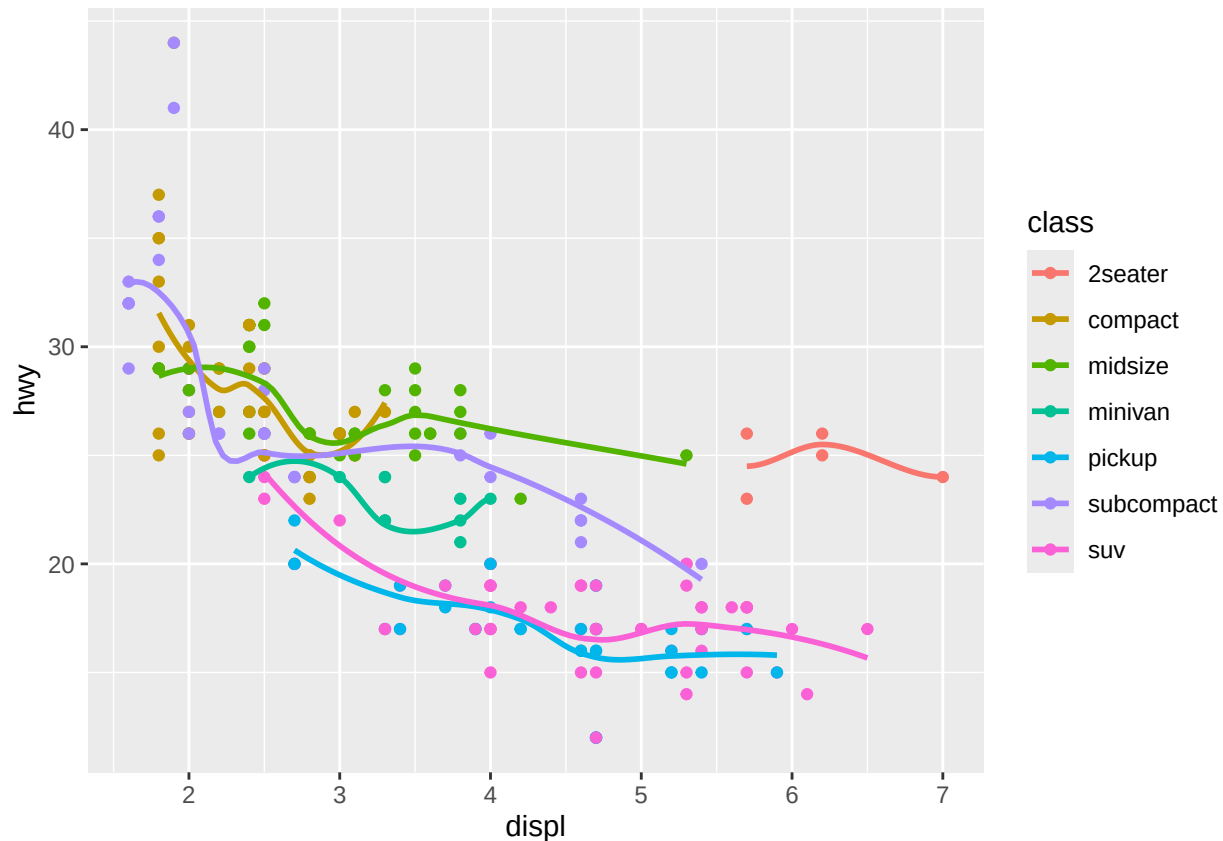
#q3

```
library(ggplot2)
# Colour mapped to class: separate colours and a legend
ggplot(mpg, aes(x = displ, y = hwy, colour = class)) +
  geom_point() +
  geom_smooth(se = FALSE)

## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = parametric,
## : span too small. fewer data values than degrees of freedom.
## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = parametric,
## : pseudoinverse used at 5.6935
## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = parametric,
## : neighborhood radius 0.5065
## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = parametric,
## : reciprocal condition number 0
## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = parametric,
## : There are other near singularities as well. 0.65044
## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = parametric,
## : pseudoinverse used at 4.008
## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = parametric,
## : neighborhood radius 0.708
## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = parametric,
```

```
## : reciprocal condition number 0
```

```
## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = parametric,  
## : There are other near singularities as well. 0.25
```



```
# Points stay one colour; smooths are grouped by class  
ggplot(mpg, aes(x = displ, y = hwy)) +  
  geom_point() +  
  geom_smooth(aes(group = class), se = FALSE)
```

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```

```
## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = parametric,  
## : span too small. fewer data values than degrees of freedom.
```

```
## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = parametric,  
## : pseudoinverse used at 5.6935
```

```
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## : reciprocal condition number 0  
  
## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = parametric,  
## : There are other near singularities as well. 0.25
```

